

Explaining the Black-Scholes Option Pricing Model

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1. Why Use Brownian Motion to Model Stock Prices?

In financial markets, asset prices fluctuate unpredictably. We use **Brownian motion**, a continuous-time stochastic process with normally distributed, independent increments to model this randomness. It is widely used to represent random movement in physics, commonly used in heat transfer simulations. However, the standard Brownian motion expression can lead to negative stock prices, which are unrealistic in financial contexts.

To resolve this, we use **geometric Brownian motion (GBM)**, which models the *percentage change* in the stock price rather than the absolute change.

Why Percentage Change Avoids Negative Prices

Modeling percentage change implies that the price follows a **multiplicative process**, not an additive one. If the change in stock price were simply added at each step, large negative changes could result in negative prices. This would occur under an additive model such as:

$$dS_t = \mu dt + \sigma dW_t$$

In contrast, geometric Brownian motion models the relative change in price:

$$dS_t = \mu S_t dt + \sigma S_t dW_t$$

Solving this gives:

$$S_t = S_0 \exp \left[\left(\mu - \frac{1}{2} \sigma^2 \right) t + \sigma W_t \right]$$

Since the exponential function is always positive, the stock price S_t will never go below zero. This reflects the real-world constraint that stock prices cannot be negative.

2. Risk-Neutral Valuation and Model Setup

In reality, investors demand higher returns for riskier assets, meaning different assets have different expected returns. However, this complicates option pricing. To simplify, we use a mathematical framework called **risk-neutral valuation**, where we assume all investors are indifferent to risk.

In this framework, we change our probability model so that all assets grow at the *risk-free rate* r instead of their actual expected return μ . This transformed world is known as the **risk-neutral measure** $\mathbb{Q}$.

Under this measure, the stock price dynamics become:

$$dS_t = rS_t dt + \sigma S_t dW_t^{\mathbb{Q}}$$

where $dW_t^{\mathbb{Q}}$ is Brownian motion under the risk-neutral measure.

In this world, the **fair price** of an option is simply the expected value of its future payoff, discounted back at the risk-free rate. Mathematically:

$$\text{Option Price} = \mathbb{E}^{\mathbb{Q}}[\text{Payoff}] \cdot e^{-rT}$$

This eliminates the need to estimate investor preferences or risk premiums, and makes option pricing more tractable and arbitrage-free.

3. Deriving the Black-Scholes PDE

Let $V(S, t)$ represent the value of an option. Using **Ito's Lemma**, we write:

$$dV = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} \right) dt + \sigma S \frac{\partial V}{\partial S} dW_t^{\mathbb{Q}}$$

We construct a portfolio $\Pi = V - \Delta S$ with $\Delta = \frac{\partial V}{\partial S}$ to eliminate randomness:

$$d\Pi = dV - \Delta dS = \left(\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

This riskless portfolio should earn the risk-free rate:

$$d\Pi = r(V - S \frac{\partial V}{\partial S}) dt$$

Equating both gives the **Black-Scholes partial differential equation (PDE)**:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

4. Boundary Conditions and Final Solution

We begin with the Black-Scholes partial differential equation:

$$\frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

where $V(S, t)$ is the value of the option, S is the stock price, t is time, σ is the volatility of the stock, and r is the risk-free interest rate.

This equation is solved under terminal conditions at maturity T :

$$\text{Call: } V(S, T) = \max(S - K, 0), \quad \text{Put: } V(S, T) = \max(K - S, 0)$$

After some algebraic manipulation the final call and put option formula's are:

$$C = SN(d_1) - Ke^{-rT}N(d_2), \quad P = Ke^{-rT}N(-d_2) - SN(-d_1)$$

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}$$

where $N(\cdot)$ is the cumulative distribution function of the standard normal distribution.

5. References

Haugh, Martin. The Black-Scholes Model. 2016.